

5.04 Linear Combinations of Random Variables			
5.04a	Linear combinations of any random variables		a) Be able to use the following results, including where $a = b = \pm 1$ and/or $c = 0$: 1. $E(aX + bY + c) = aE(X) + bE(Y) + c$ 2. if X and Y are independent then $\text{Var}(aX + bY + c) = a^2 \text{Var}(X) + b^2 \text{Var}(Y)$
5.04b	Linear combinations of normal random variables		b) Be able to use the following results: 1. if X has a normal distribution then $aX + b$ has a normal distribution, 2. if X and Y have independent normal distributions then $aX + bY + c$ has a normal distribution.